Sem (IX) Tutorial (II) Date:9.9.2026

1. Consider the random variable X with probabilities $P = (\frac{3}{10}, \frac{1}{4}, \frac{1}{5}, \frac{1}{10}, \frac{3}{40}, \frac{1}{40})$.
   (a) Construct a binary Huffman code for this random variable. What is its expected length?
   (b) Construct a ternary Huffman code for this random variable. What is its expected length?

2. Consider the random variable X with probabilities $P = (0.3, 0.25, 0.2, 0.1, 0.1, 0.05)$.
   Find a Shannon-Fano-Elias code and Shannon code for X. What are the average codeword length for these encoding.

3. Consider the code {10, 00, 11, 110}.
   (i) Is it uniquely decodable? (ii)Is it instantaneous? (iii)Is it singular?

4. Which of the following codeword lengths can be the word lengths of a 3-ary Huffman code, and which cannot?
   (i) (1, 2, 3, 5, 5, 5, 5)
   (ii) (1, 1, 2, 2, 3, 3, 3)
   (iii) (1, 1, 2, 2, 2)